

NAKHON SAWAN, Thailand

WMO: 484000

Lat: 15.6717N

Lon: 100.1322E

Elev: 35

StdP: 100.90

Time Zone: 7.00 (E07)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 12	(b) 16.3	(c) 17.9	(d) 11.1	(e) 8.3	(f) 22.6	(g) 12.8	(h) 9.3	(i) 23.1	(j) 4.5	(k) 28.3	(l) 3.9	(m) 28.4	(n) 0.1	(o) 40	(p) 0.336

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 4	(b) 10.5	(c) 39.0	(d) 26.4	(e) 37.9	(f) 26.1	(g) 36.8	(h) 26.0	(i) 28.3	(j) 34.2	(k) 28.0	(l) 33.8	(m) 27.7	(n) 33.3	(o) 1.6	(p) 180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
27.0	22.8	31.0	26.6	22.3	30.6	26.3	21.9	30.3	91.5	34.4	89.9	34.0	88.6	33.5	30.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years						
Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max					
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
4.9	4.0	3.4	13.8	40.3	2.1	1.1	12.2	41.1	11.0	41.7	9.8	42.4	8.2	43.2		
			12.3	29.1	2.0	0.5	10.8	29.4	9.6	29.7	8.4	30.0	6.9	30.3		
			DB													
			WB													

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	28.9	26.1	28.6	30.7	31.9	30.8	30.1	29.4	29.2	28.8	28.6	27.5	25.7
	DBStd	2.50	2.27	2.29	2.14	1.97	1.93	1.14	1.14	0.97	0.99	1.15	1.76	2.38
	HDD10.0	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD18.3	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDD10.0	6915	499	519	641	657	646	603	602	595	564	576	524	488
	CDD18.3	3873	241	286	383	407	387	353	344	337	314	317	274	229
	CDH23.3	46133	2394	3378	5124	5792	5078	4394	4090	3877	3475	3517	2843	2171
CDH26.7	22359	1061	1724	2887	3501	2722	2109	1812	1634	1387	1427	1194	902	
Wind	WSAvg	0.8	0.6	1.0	1.4	1.3	0.9	0.9	0.8	0.7	0.4	0.3	0.4	0.5
Precipitation	PrecAvg	1165	5	12	40	65	156	134	141	173	249	160	23	5
	PrecMax	1558	25	68	170	196	334	255	255	366	377	557	127	50
	PrecMin	634	0	1	0	4	35	30	46	67	76	1	0	0
	PrecStd	218	8	18	41	54	76	55	53	68	76	100	28	10
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	35.1	37.4	39.1	41.0	40.3	37.3	36.2	35.3	34.8	34.7	35.1	34.7
		MCWB	24.2	24.2	25.2	27.0	27.3	26.8	27.0	26.9	27.5	26.9	26.0	24.9
	2%	DB	34.1	36.2	37.9	39.5	38.2	35.7	34.8	34.3	33.9	33.7	34.0	33.5
		MCWB	24.0	24.6	25.1	26.5	27.0	26.9	27.0	26.9	27.3	26.7	25.8	24.6
	5%	DB	33.1	35.2	37.0	38.4	36.7	34.7	33.9	33.5	33.1	33.1	33.1	32.6
		MCWB	23.5	24.6	25.2	26.5	26.9	27.0	26.8	26.8	27.1	26.5	25.4	24.1
	10%	DB	32.0	34.1	36.0	37.2	35.4	33.7	33.0	32.6	32.2	32.3	32.1	31.4
		MCWB	23.1	24.3	25.2	26.4	27.0	27.0	26.7	26.7	26.9	26.4	25.1	23.5
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	26.6	27.2	27.6	28.6	28.8	28.7	28.2	28.2	28.4	28.2	27.8	26.8
		MCDB	32.1	34.1	35.3	36.8	35.5	33.8	33.4	32.9	33.1	32.8	32.7	32.1
	2%	WB	25.8	26.5	27.1	28.0	28.2	28.1	27.7	27.6	27.8	27.7	27.1	25.7
		MCDB	31.1	33.3	34.7	35.9	34.8	33.3	32.8	32.4	32.4	32.1	31.8	31.0
	5%	WB	25.0	26.0	26.6	27.6	27.7	27.7	27.3	27.3	27.5	27.3	26.6	25.1
		MCDB	30.3	32.5	33.8	35.1	34.1	32.8	32.3	32.0	31.8	31.5	31.0	30.3
	10%	WB	24.4	25.5	26.2	27.2	27.4	27.2	27.0	27.0	27.2	26.9	26.0	24.4
		MCDB	29.4	31.4	32.9	34.4	33.4	32.4	31.9	31.6	31.3	30.9	30.1	29.4
Mean Daily Temperature Range	5% DB	MDBR	12.0	11.8	10.9	10.5	8.8	7.9	7.5	7.4	7.3	7.6	9.1	11.0
		MCDBR	12.9	12.2	12.0	11.8	10.6	9.2	8.7	8.5	8.4	8.6	10.0	11.5
		MCWBR	4.3	3.4	3.2	3.1	2.7	2.8	2.9	2.9	3.0	2.7	3.0	3.7
	5% WB	MCDBR	10.3	10.7	10.8	10.6	9.3	8.3	8.0	7.9	7.8	7.7	8.6	10.1
		MCWBR	3.8	3.3	3.2	3.2	3.1	3.0	3.0	3.0	3.0	2.9	3.0	3.4
Clear-Sky Solar Irradiance	taub	0.507	0.595	0.672	0.602	0.497	0.442	0.434	0.438	0.476	0.561	0.495	0.498	
	taud	2.010	1.821	1.679	1.877	2.219	2.382	2.405	2.379	2.260	1.973	2.132	2.050	
	Ebn at Noon	781	731	690	741	811	848	857	861	832	749	788	775	
	Edh at Noon	169	213	252	207	145	122	119	124	139	181	149	159	
All-Sky Solar Radiation	RadAvg	4.76	5.23	5.47	5.97	5.62	5.24	4.93	4.82	4.76	4.76	4.82	4.65	
	RadStd	0.28	0.21	0.34	0.28	0.35	0.32	0.29	0.28	0.25	0.28	0.26	0.17	

Historical Trends

DBAvg	Heating			Cooling			Degree-Days			
	99% DB	99% DP	1% DB	1% WB	1% DP	1% WB	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)
(46) Station Only	N/A	N/A	N/A	N/A	+0.42	+0.40	N/A	N/A	+100	+100
(47) Regional (22 neighbors)	N/A	N/A	N/A	N/A	+0.20	+0.25	N/A	N/A	N/A	N/A

Nomenclature: See separate page